Feasibility Memo: ED Triage Dataset

Is the yaleemmlc_admissionprediction_triage.csv dataset suitable for building a triage prediction model?

To: Dr. De Fretias & the ED Board
From: Skye Thomas
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Re: Feasibility Assessment for an AI-Assisted ED Triage Model

Verdict

The dataset is suitable for developing a baseline triage prediction model in Week 6. However, several limitations should be addressed during data preparation and model evaluation to ensure the model performs reliably and safely in a clinical setting.

Dataset Summary

The dataset contains 55,121 emergency department (ED) encounters collected from three departments. Each patient record includes 225 variables, consisting of 25 structured features such as demographics, arrival information, and triage vital signs, along with 200 binary chief complaint indicators.

The prediction target is the Emergency Severity Index (ESI), which classifies patients into five levels of urgency, with ESI-1 representing the most critical patients and ESI-5 the least urgent.

Patients range in age from 18 to 107 years, with a median age of 55 years. Approximately 39% have Medicaid insurance, 32% have Medicare, and 25% have commercial insurance. Around one-third of patients arrive by ambulance, while most of the remaining patients either arrive by private vehicle or walk into the emergency department. The data comes from a single U.S. healthcare system and should not be assumed to represent other hospitals or healthcare settings.

One encouraging finding from the initial data profiling is that all 25 structured variables are complete, with no missing values in demographics, vital signs, or the ESI outcome. Although this simplifies data preparation, complete electronic health record data is uncommon, so it would be worthwhile to confirm how the dataset was processed before being released.

Top Three Data Quality Concerns

1. Severe Class Imbalance

The most significant issue is the imbalance in the ESI classes. Nearly half of all patients are classified as ESI-3, whereas only 77 patients belong to ESI-1, representing approximately 0.1% of the dataset. As a result, a model trained without addressing this imbalance could achieve high overall accuracy while performing poorly on the most clinically important cases:

ESI level	Patients	% of total
1 (most urgent)	77	0.1%
2	17,924	32.5%
3	27,010	49.0%

4	8,896	16.1%
5 (least urgent)	1,214	2.2%

Further analysis shows that ESI-1 patients are mainly identified by a small number of protocol-triggered chief complaints rather than by extreme vital signs. Complaints such as stroke alert, respiratory distress, hypotension, and unresponsive patients occur much more frequently among ESI-1 patients than in the rest of the dataset. In comparison, average heart rate and oxygen saturation differ only slightly from the overall patient population.

Arrival by ambulance is another strong indicator of ESI-1 status, with approximately 77% of ESI-1 patients arriving by ambulance compared with about 34% of all patients.

These complaint variables do not appear among the highest-ranked correlation features because they occur in very few patients overall. While they contribute little to population-wide correlation measures, they remain highly informative for identifying the sickest patients. For this reason, these complaint flags, along with ambulance arrival, should be included as engineered features in the Week 6 model and evaluated using ESI-1 recall rather than overall accuracy alone.

Generating synthetic ESI-1 patients is not recommended. With only 77 real cases spread across nearly 200 binary complaint variables, synthetic data could produce unrealistic combinations of symptoms and vital signs that are unlikely to occur in real clinical practice. A safer approach is to use class weighting, stratified data splitting, and evaluation metrics that prioritize correctly identifying ESI-1 patients while relying only on real patient records.

A final limitation is the small sample size for several complaint categories. Some complaints occur in only three or four ESI-1 patients, meaning the percentages could change considerably if only one patient were added or removed. While the overall pattern is reliable, the exact percentages should be interpreted with caution.

2. Sparse Chief Complaint Variables

Many of the chief complaint variables occur very rarely. Approximately 75% of the 200 complaint indicators are present in fewer than 0.5% of patients, meaning many variables contribute little information during model training.

Additionally, nine complaint variables contain the value "2" instead of the expected binary values of 0 or 1. This most likely represents a data entry issue rather than a valid third category and should be corrected before model development.

3. Limited Representation

The dataset also has limitations in terms of representation. More than half of the patients are White, while several racial groups contain relatively few patients. As a result, the model may have limited ability to learn meaningful patterns for these smaller groups.

The analysis also identified differences in ESI assignment between Black and White patients. Black patients were assigned ESI-4 more frequently and ESI-2 less frequently than White patients. The available data cannot determine whether these differences reflect genuine differences in patient presentation or existing triage practices. Regardless of the cause, any model trained on this dataset may learn these patterns, making fairness evaluation an important part of model development.

Reasons to Proceed

Despite these limitations, the dataset provides a strong foundation for developing a baseline triage prediction model.

First, the most predictive variables are consistent with clinical expectations. Features such as ambulance arrival, age, oxygen saturation, chest pain, and shortness of breath are all factors that healthcare professionals commonly consider during triage, increasing confidence that the dataset captures meaningful clinical information.

Second, the dataset contains more than 55,000 patient encounters, providing enough observations to train, validate, and test a machine learning model while still maintaining a separate evaluation dataset.

Finally, the identified data quality issues are manageable. Problems such as unrealistic glucose values, incorrect complaint coding, and temperature units can all be addressed through standard preprocessing techniques before model training.

Feature Shortlist

Ranked by clinical intuition combined with simple correlation against ESI (computed directly from the dataset):

Rank	Feature	Justification
1	Arrival by ambulance	EMS crews already perform a field severity assessment before the patient reaches the ED — the strongest single predictor of acuity in this dataset.
2	Age	Older patients carry more comorbidity and less physiologic reserve, increasing acuity for the same presenting complaint.
3	Oxygen saturation	Low oxygen saturation is a recognized danger-zone vital sign that should prompt urgent evaluation regardless of other findings.
4	Chest pain	Carries a life-threatening differential (e.g. cardiac or pulmonary causes) that clinicians triage up by default.
5	Shortness of breath	A second danger-zone complaint, independent of the vital-sign values it may or may not have triggered yet.
6	Suicidal ideation	Psychiatric emergencies follow a distinct, urgent safety pathway regardless of physical vital signs.
7	Altered mental status	A nonspecific but reliable marker of an active neurological or systemic emergency.
8	Respiratory rate	Has a well-established, direct link to physiologic decompensation.
9	Heart rate	Tachycardia is an early, if nonspecific, signal of hemodynamic stress.
10	Glucose	Extreme values flag metabolic emergencies such as DKA or severe hypoglycemia that can otherwise look deceptively stable.

Recommendations

- Exclude the variables disposition and previousdispo, because they become available only after triage and would introduce data leakage.
- Stratify training and testing datasets by ESI class to preserve the original class distribution.

- Evaluate model performance using recall and other class-sensitive metrics instead of overall accuracy alone, because ESI-1 patients are extremely rare.
- Document the temperature variable clearly as being recorded in Fahrenheit to avoid unit conversion errors.
- Monitor model performance across demographic groups to identify any fairness concerns before considering clinical implementation.

Conclusion

Overall, the exploratory analysis indicates that the dataset is suitable for developing a baseline emergency department triage prediction model. Although challenges such as severe class imbalance, sparse complaint variables, and limited representation are present, these issues are manageable using appropriate preprocessing and evaluation techniques. With these considerations incorporated into the modelling process, the dataset provides a strong basis for continuing to Week 6 model development.